

# Expanding Double Brackets

AQA KS3 · Year 8 · Foundation

To expand double brackets, multiply every term in the first bracket by every term in the second bracket. You can use the FOIL method (First, Outside, Inside, Last) or the grid method to organise your working. Then collect like terms to simplify.

## Method

- 1 Multiply each term in the first bracket by each term in the second
- 2 Write each product in its grid cell
- 3 Collect like terms (the two middle terms)
- 4 Write the simplified answer
- 5 Check by substituting a value

## Common mistakes

- Forgetting the last term: you must multiply the constant in the first bracket by the constant in the second.
- Sign errors with negatives: remember that a negative times a negative gives a positive.
- Writing  $x + x$  instead of  $2x$  for the middle term when collecting like terms.
- Forgetting that  $x \times x = x^2$ , not  $2x$ .

## Worked example

Expand and simplify:  $(x + 2)(x + 3)$

Step 1 — Draw the grid with  $(x + 2)$  down the side and  $(x + 3)$  along the top.

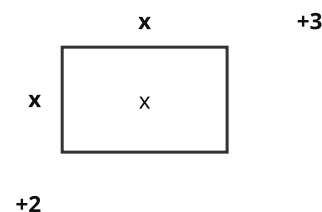
Step 2 — Fill in:  $x \cdot x = x^2$ ,  $x \cdot 3 = 3x$ ,  $2 \cdot x = 2x$ ,  $2 \cdot 3 = 6$

Step 3 — Collect like terms:  $3x + 2x = 5x$

Step 4 — Write the answer:  $x^2 + 5x + 6$

Check ( $x = 10$ ):  $(10+2)(10+3) = 12 \times 13 = 156$   
and  $100 + 50 + 6 = 156$

## Visual aid — grid method



2  
 $3 \times 2 \times 6$

Each cell shows the product of its row and column headers:  $(x+2)(x+3) = x^2 + 5x + 6$ .

# Questions

Expand and simplify — show your working in your workbook

**Q1 of 8**

[1] ● ○ ○ ○ ○

Expand and simplify:  $(x + 1)(x + 2)$

**Q2 of 8**

[1] ● ○ ○ ○ ○

Expand and simplify:  $(x + 3)(x + 4)$

**Q3 of 8**

[2] ● ● ○ ○ ○

Expand and simplify:  $(x + 5)(x - 2)$

**Q4 of 8**

[2] ● ● ○ ○ ○

Expand and simplify:  $(x - 3)(x + 4)$

**Q5 of 8**

[2] ● ● ● ○ ○

Expand and simplify:  $(x - 2)(x - 5)$

**Q6 of 8**

[3] ● ● ● ○ ○

Expand and simplify:  $(2x + 1)(x + 3)$

**Q7 of 8**

[3] ● ● ● ● ○

Expand and simplify:  $(3x - 2)(x + 4)$

**Q8 of 8**

[3] ● ● ● ● ●

Expand and simplify:  $(2x + 3)(3x - 1)$